

Abstract No.

Project HOPE: a longitudinal multidimensional profiling of high-grade gliomas in children, adolescents and young adults

Sina Neyazi^{1,2,}, Li Jiang¹, Shashank Katiyar¹, Sara Danielli¹, Olivia Hack¹, Costaza Lo Cascio¹, Andrezza Nascimento¹, Cuong Nguyen¹, Jacob Rozowsky¹, Ilon Liu¹, Owen Hoare³, Karin Shamardani³, Kennedy Cunliffe-Koehler⁴, Johannes Gojo⁵, Alexander Beck⁶, Ulrich Schüller^{2,7}, Jacquelyn Jones^{8,9}, Erin White^{8,9}, Olivia Vizzini^{8,9}, Keith Ligon^{8,9}, Lissa Baird⁹, Sanda Alexandrescu⁹, Jennifer Cotter¹⁰, Michael Prados³, Adam Resnick¹¹, Lin Wang¹², Michelle Monje⁴, Aaron Diaz³, and Mariella G. Filbin¹*

¹Dana-Farber Boston Children's Cancer and Blood Disorders Center

²University Medical Center Hamburg-Eppendorf

³University of California, San Francisco

⁴Stanford University

⁵Comprehensive Center for Pediatrics and Comprehensive Cancer Center, Medical University of Vienna

⁶Ludwig-Maximilians-University Munich

⁷Research Institute Children's Cancer Center Hamburg

⁸Dana-Farber Cancer Institute

⁹Boston Children's Hospital

¹⁰Children's Hospital Los Angeles

¹¹Children's Hospital of Philadelphia

¹²Chinese Academy of Medical Sciences & Peking Union Medical College

*Presenting author

Pediatric-type diffuse high-grade gliomas (pHGGs) are aggressive central nervous system tumors that are the leading cause of cancer-related death in children. pHGGs comprise histologically, genetically, and molecularly distinct subtypes with characteristic driver mutations, anatomical localizations, and ages of onset. Although pHGGs share many features with adult gliomas, they represent clinically and biologically distinct tumor entities. Despite growing insights into the cellular architectures of pHGGs, the evolution of malignant cell states and tumor microenvironments during tumor progression remains largely unexplored. In this study, we generated an extensive longitudinal single-nucleus and spatial transcriptomic atlas of pediatric high-grade gliomas (pHGG) from children, adolescents and young adults using single-nucleus RNA sequencing and Xenium In Situ spatial transcriptomics. Our cohort includes 49 paired HGG samples across tumor subtypes, spanning diffuse midline glioma (DMG), IDH-mutant (IDH-HGG) and H3- and IDH-wildtype high-grade glioma (HGG), representing the most comprehensive longitudinal pediatric glioma cohort profiled to date. Single-nucleus profiling uncovered shared and distinct cellular subpopulations in different pHGG subtypes. Despite notable similarities in transcriptional profiles of primary and recurrent pHGGs, recurrent HGGs showed increased proportions of malignant cells with a hypoxia-associated program, coupled with a reduction in malignant glial-like cell populations. Spatial transcriptomic analysis of 71 tumor sections from 31 pHGGs revealed subtype-specific global spatial organizations and localized cellular niches. Upon analyzing the density of tissue regions, we found that the normal cells occur in less dense regions than malignant cells; within the malignant compartments, we found OPC-like cells consistently existed in denser regions than AC-like cells. Overall, our study characterizes the cellular landscape and spatial architecture across pHGG subtypes and provides a detailed delineation of the spatiotemporal dynamics during progression, paving the way for the identification of novel therapeutic targets.



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